

John Feczko

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Web Portfolio: <https://john-feczko.github.io/>

Education

Villanova University – College of Engineering, Villanova, PA 2027
Bachelor of Science in Mechanical Engineering, Minor in AI, Concentration in Dynamic Systems, Minor in Humanities
GPA: 3.95 Dean's List, all semesters

Senior Research; AI Robot Guide Fall 2026

- This project will study how AI acting through a mobile robotic system can interpret environmental information and generate motion decisions using a combination of sensors, embedded computing hardware, and AI-based software tools
- Research tools and techniques will include the use of an NVIDIA DGX Spark, Jetson Orin boxes, APIs, and Optical Character Recognition (OCR), along with GPS and camera data

Mechanical Engineering Sophomore Research Scholar; Smart Textiles 2024

- Researched nanoparticle assembly and atomic monolayer analysis, including the enhanced detection of PFAS particles and fireproofing of ordinary textiles through nanoparticle coating

Class Projects; Hands-on design projects within coursework, involving CAD, programming, prototyping, and fabrication, including:

- Built a functioning web app using generative AI orchestration, user GPS data, and RAG documentation to make informed responses on the history of the user's location
- Designed a cost-effective automatic fire suppressant system intended for arid-climate refugee tent camps
- Conducted full lifecycle development of a small robotic "Beetle-Bot" system from concept to fabrication, electrical integration, testing, and post-failure analysis

Work Experience

Arconic Corporation; Manufacturing Engineering Internship Summer 2026

- Work on projects for the senior process engineer at an aluminum processing plant to improve plant operating efficiency
- Investigated wastewater treatment system and identified design improvements estimated to save up to \$1,000,000 per year
- Analyzed the current natural gas infrastructure and provided recommendations on optimal meter positions

Handshake AI Fellowship; AI Model Trainer 2025 – Present

- Conduct AI feedback research as an independent contractor with industry AI development companies
- Develop and evaluate domain-specific prompts to assess the performance of large language models (LLMs)
- Analyze LLM outputs for scientific accuracy, clarity, and depth in specialized subfields

BCI Engineering; Summer Intern, Leetsdale, PA 2023 – 2024

- Designed parts, drawings, assemblies, Bill of Materials (BOM) and applied General Drafting & Tolerancing (GD&T) in CAD
- Conducted rapid prototyping for new tools and models to improve quality assurance efficiency
- Created Quality Assurance reviews using First Article Inspection (FAI) reports and load-test data summary reports

Activities & Leadership

Baja SAE Offroad Vehicle Steer Team Lead	2026 – 2027
American Society of Mechanical Engineers (ASME)	2023 – Present
Physics II Tutor	2025 – Present
Faith & Reason Scholar Program	2024 – Present
Engineering service trip to Kenya	2025
Villanova Men's Rowing	2023 – 2025

Technical Skills

- Claude Code, prompt engineering, Retrieval Augmented Generation (RAG), hosting AI agents, MCP tools, model finetuning
- SolidWorks, including simple Finite Element Analysis (FEA) in mechanical stress and heat transfer simulations
- Autodesk Inventor, including the creation of parts, assemblies, drawings, and BOMs
- AutoCAD, Fusion, FreeCAD, MATLAB, Maple, Python, Arduino, and C++
- Advanced with standard and resin 3D printing, laser cutting/engraving, and traditional machining (lathe, mill, saw, etc.)
- Experience with CNC milling, CNC lathing, and WASER waterjet cutting